OOS Impact Analytics — Methodology Sheet

How the dashboard detects out-of-stock and demand-throttling events and values their revenue / contribution-margin impact. Worked example: **VV-VITA-216 — Kudzu Plus Capsules**, analysed EU-wide (Pan-EU). Prepared for review.

1. Data sources

• **Amazon FBA Inventory Ledger** — real daily sellable warehouse balance and units shipped per SKU. The account runs **Pan-EU**, so stock is pooled: a SKU is physically out only when the whole EU balance hits zero.
• **Novadata margin export** — daily units, sales and contribution margin (CM3) per SKU and marketplace; supplies price and CM3 per unit for valuation.
• **Shopify catalog** — English product titles.

2. Two core measures

Demand rate (lambda). Trailing 90-day average units per calendar day, per SKU and marketplace. Daily sales follow an approximate Poisson process, so the chance of selling zero on a normal day is P(0)=e^(-lambda): ~37% at 1/day, ~5% at 3/day, ~0.7% at 5/day. A zero-sales day is therefore only treated as a stock-out when lambda is high enough that zero is a real anomaly (default threshold 3/day).
Reach (days-of-supply). EU sellable stock divided by trailing 28-day average units shipped — how many days of cover remain.

3. Daily classification (one label per SKU x marketplace x day)

Evaluated in priority order; the first match wins:

Priority	Label	Condition	Counts as
1	Physical OOS	EU sellable balance = 0	Lost (involuntary)
2	Critically low	Reach < 3 days	Lost (involuntary)
3	Cooling down	3 <= reach < 30 days, still selling (units > 0), throttled (price >= +8% and/or ad cut >= 50%)	Miss (voluntary)
4	Demand gap (EU)	Units = 0 on a day enclosed by sales, lambda >= 3/day (incl. throttles that hit 0)	Lost (involuntary)

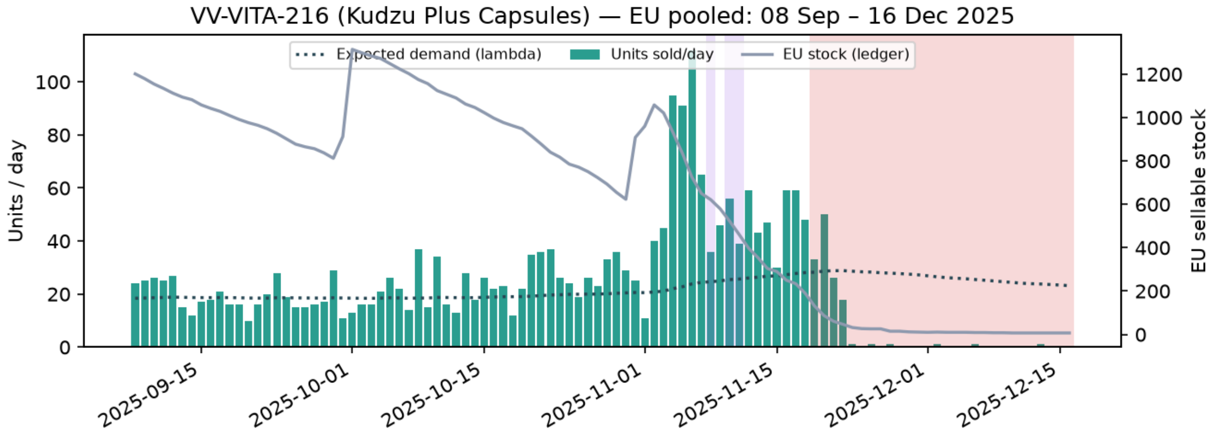
Guards: the SKU must already be selling (ignores pre-launch); a zero-run must be enclosed by later sales or be within the last 21 days (ignores discontinued tails). Cooling-down separates deliberate throttling — which otherwise looks like a demand drop — from genuine stock-outs, so the two are never double-counted.

4. Impact valuation

On any flagged day: **missed units = max(lambda - actual units, 0)** (the expected demand we didn't capture). Valued at the SKU's trailing average selling price and CM3 per unit:
Lost / Miss revenue = missed units x avg price
Lost / Miss CM3 = missed units x avg CM3 per unit (the true P&L impact).
'Lost' = involuntary (OOS); 'Miss' = voluntary (cooling down). Consecutive flagged days are grouped into discrete events.

5. Worked example

VV-VITA-216 — Kudzu Plus Capsules, EU-wide (Pan-EU pooled). Full year: **114 OOS days** (lost €59.538 revenue / €18.646 CM3) and **4 cooling-down days** (miss €785 revenue / €214 CM3).



Highlighted window 08 Sep 2025 – 16 Dec 2025: EU stock ranged from ~1.314 down to ~7 units and reach bottomed at ~0 day(s). In this window the model flagged **28 OOS day(s)** (red/orange shading) and **3 cooling-down day(s)** (purple) — the SKU was throttled (price up / ad spend cut) while stock was tight, then went into genuine stock-out as reach collapsed. Note the balance rarely hits literally zero, so the **reach < 3** rule, not a balance-of-zero rule, is what catches the hard stock-out. Stock was replenished on **02 Dec** (EU balance jumps in the chart), after which sales recover and the flags clear — the table's last rows show those first restock days.

What "cooling down" means here. On the purple days the SKU kept selling (units > 0) but demand was deliberately damped to stretch the remaining stock to the next shipment. **Before cooling down** it sold at ~€21.50 with ~€0/day of ad spend; **during the throttle** the price was lifted to ~€25.01 and ad spend cut to ~€0/day. Sales fell to about 39/day versus the expected ~20/day, and that gap is booked as a **voluntary miss** — a pricing / advertising choice to protect availability and the listing's ranking — not a lost stock-out. Crucially the SKU keeps selling: the moment a throttle pushes sales to zero it is no longer cooling down and counts as OOS (lost). Cooling down trades a little volume now (the miss) to avoid the larger ranking and lost-sales hit of a full stock-out.

Representative days (Lost/Miss shown as CM3 €):

Date	Units	λ	EU stk	Reach	Price	Category	Lost €	Miss €
11-08	36	25	621	18	25.01	Cooling down		
11-10	56	25	522	14	24.94	Cooling down		
11-11	39	26	461	12	25.55	Cooling down		
11-19	33	28	131	3	23.67	Critically low (<4d)		
11-20	50	29	87	2	24.80	Critically low (<4d)		
11-21	26	29	61	1	25.48	Critically low (<4d)	21	
11-22	18	29	47	1	24.79	Critically low (<4d)	82	
11-23	1	29	32	1	19.90	Critically low (<4d)	208	
11-24	0	28	27	1	–	Critically low (<4d)	213	
12-01	0	27	10	0	–	Critically low (<4d)	203	
12-02	1	26	11	0	25.90	Critically low (<4d)	193	
12-03	0	26	10	0	–	Critically low (<4d)	198	

6. Caveats & rules

This sheet illustrates the CORE rules on one example; the interactive dashboard is the source of truth and includes further refinements (blocked-listing rule, promo-elevated counterfactual, episode bridging, region-specific reach EU 4 / GB 12).

Available stock = on-hand sellable + units in transit between FCs, so a Pan-EU redistribution (e.g. a bulk receipt landing at the Poland hub, then moving to other FCs over ~1-3 weeks) is not read as a stock-out. **Customer returns do not end a stock-out** — once OOS, a SKU stays OOS through return-driven balance blips until a genuine inbound Receipt or sales recover to >= half of expected. **GB** is a separate (non-Pan-EU) warehouse, analysed on its own.

Impact figures are estimates (assume the SKU would have sold at its recent rate). Ad-spend-cut detection only applies from when Novadata began reporting Advertising Costs (~Feb 2026); the price-hike lever spans the full year. Reach is EU-pooled, so reach-based rules need ledger coverage for the SKU. All thresholds (3/day, reach 4 & 30, +15%, -70%) are tunable in the dashboard.